

## Hypothesis two sample t-tests



- 1
  - a  $p = 0.73 > \alpha$ , therefore do not reject the null hypothesis.
  - b  $p = 0.74 > \alpha$ , therefore do not reject the null hypothesis.
  - c  $p = 0.14 < \alpha$ , therefore reject the null hypothesis.
  
- 2
  - a  $\mu_1$  is mean hours of study for Year 11 students.  
 $\mu_2$  is mean hours of study for Year 12 students.  
 $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 < \mu_2$
  - b One tailed test
  - c  $p = 0.000342$
  - d As  $p < \alpha$  we say that there is enough evidence to reject  $H_0$  on a 1% level of significance. Therefore we can comment that Year 11 students do appear to study less than Year 12 students.
  
- 3
  - a  $\mu_1$  is mean bulb hours for the old globes.  $\mu_2$  is mean bulb hours for the new globes.  
 $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 < \mu_2$
  - b One tailed test
  - c  $p = 0.04$
  - d As  $p < \alpha$  we say that there is enough evidence to reject  $H_0$  on a 5% level of significance. Therefore we can comment that the new globes do appear to last longer than the old globes.
  
- 4
  - a The population data is normally distributed and the two sets of data have equal standard deviations (or variances).
  - b  $\mu_1$  is mean deaths using poison A.  $\mu_2$  is mean deaths using poison B.  
 $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 \neq \mu_2$
  - c  $p = 0.16$
  - d As  $p > \alpha$  we do not reject the null hypothesis. There is no evidence that poison A is better or worse than poison B.
  
- 5
  - a  $H_0 : \mu_A = \mu_B, H_1 : \mu_A \neq \mu_B$
  - b Two tailed test
  - c  $p = 0.077$
  - d As  $p > \alpha$  we do not reject the null hypothesis that the drugs perform the same and there is no statistically significant difference in the sleep outcomes.

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- a  $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 \neq \mu_2$
- b Two tailed test
- c  $p = 0.01$
- d As  $p < \alpha$  we reject the null hypothesis that the average scores are the same for both groups and we can say that there is a statistically significant difference between the test scores of the males compared to the females.
- e There is a 10% chance of incorrectly rejecting  $H_0$ .

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- a  $H_0 : \mu_1 = \mu_2, H_1 : \mu_1 > \mu_2$
- b  $p = 0.53$  therefore  $p > \alpha$  and we do not reject the hypothesis that native speakers score higher than non-native speakers on the reading test.